

Concurrency + Kafka + CompletableFuture Summary

1. Concurrency

Concurrency means executing multiple tasks in parallel to improve performance and reduce latency.

2. CompletableFuture

Used for parallel execution inside a service. Example: fraud check, payment, and audit run together.

Example:

```
CompletableFuture.allOf(fraudCheck, paymentProcess, auditLog).join();
```

Explanation:

- allOf(): waits for all tasks
- join(): blocks until all complete
- paymentProcess.join(): gets result

3. Kafka Usage

Kafka is used for communication between microservices. Example: sending payment-success or payment-failed events.

Example Flow:

Order Service → Kafka → Payment Service → Kafka → Inventory Service

4. Combined Design

Kafka handles distributed async communication, CompletableFuture handles parallel execution inside service.

5. Improved Code Example:

```
boolean fraud = fraudCheck.join();  
boolean payment = paymentProcess.join();  
if (fraud && payment) send success else failure
```

6. Key Benefits

- Faster processing
- Scalable system
- Better resource utilization

7. Interview Tip

Use Kafka for inter-service async and CompletableFuture for intra-service parallelism.